ELECTRICIAN REFERENCE

Drilling | Cavity Depths | Conduits | Switch Boards

PROJECT

Owner : Ganesh Prasad D
Location : Chitradurga, Karnataka
House type : 2-floor residential, North-facing main entrance
GF ceiling : 11 ft FF ceiling : 10 ft
DB location : West wall, foyer, behind door swing
DB size : 48-way Schneider Acti9 IEF48
Document : v1.0 - 2026-05-06

READ THIS FIRST

1. THIS DOCUMENT REPLACES SITE INSTRUCTIONS. If anything on site differs from this PDF, STOP and call before chasing or drilling.

2. EVERY SMART-SWITCH BOARD GETS A 65 mm DEEP BOX (not 50 mm). The Sonoff/Aqara relay module sits behind the switch plate; 50 mm is too shallow.

3. EVERY SWITCH BOARD GETS A NEUTRAL WIRE (black). No exceptions, even on dumb switch zones, because we may convert them to smart later.

4. EARTH WIRE (green/yellow) reaches every switch, socket, and metal fitting. Test continuity before plastering.

5. Power conduits and Cat6 conduits are NEVER in the same pipe. Always separate.

1. The 5 Rules That Apply Everywhere

>> Rule 1 - NEUTRAL EVERYWHERE

Every switch board (smart or dumb) receives a NEUTRAL wire (black). Smart relay modules need it. Without neutral, the smart switch goes dead. Cap unused neutrals at the box; do not omit them.

>> Rule 2 - 65 mm DEEP BOXES AT SMART SWITCHES

All smart-switch locations get a 65 mm-deep GI MS box, NOT the standard 50 mm. Sizes: 1-mod 75x75x65, 2-mod 130x75x65, 3-mod 175x75x65, 4-mod 230x75x65. Sonoff ZBMINI R2 / Aqara T1 relay sits behind the switch plate; 50 mm cannot fit module + bent wires + terminals.

>> Rule 3 - STANDARD HEIGHTS (FFL = Finished Floor Level)

Switch boards: 1200 mm (centre of plate)
Geyser switches: 1050 mm
PIR bathroom switches: 1200 mm (outside door, handle side)
Sockets (general): 300 mm
Bedside sockets: 600 mm
Counter sockets (kitchen): 1100 mm
AC sockets: 1850 mm
Geyser outlet (in attic): 1850 mm
Wi-Fi router/AP wall plate (FF Living): 2400 mm

>> Rule 4 - EARTH IS NOT OPTIONAL

Green/yellow earth wire reaches every switch, every socket, every metal fitting. Test earth continuity at every metallic body with clamp meter (< 1 ohm) before plastering.

>> Rule 5 - POWER AND DATA NEVER IN THE SAME PIPE

Cat6 picks up interference from 230V lines. ALWAYS use separate conduits: 25mm grey (LV-25) for Cat6 / network / camera data; 25mm red/blue for power.

Wire Colour Code (Indian Standard - Mandatory)

Colour	Function	Where used
RED	LIVE (phase) - 230V hot	Every live wire from MCB outward
BLACK	NEUTRAL - return	Every switch board, every socket
GREEN/YELLOW	EARTH	Every switch, socket, metal fitting
BLUE	Switched LIVE leg	Between switch and the load (light)

CRITICAL

A black wire with current is ALWAYS NEUTRAL. A blue wire is the SWITCHED LIVE leg. Do not confuse them.

2. Conduit Colour Code

Use coloured PVC conduit where available. If only grey is available locally, wrap each conduit end in coloured insulation tape and label with permanent marker BEFORE plastering. Once plastering is done, the conduit is invisible - this is your only chance to mark it.

Colour	Size	Carries	Wire inside
RED	25 mm	Lighting power	1.5 sqmm 3-core
BLUE	25 mm	Sockets / AC / Geyser / Hob power	2.5 or 4 sqmm 3-core
GREY	25 mm (LV-25)	Cat6 / network / camera data	1 or 2 Cat6 UTP
GREY	16 mm (LV-16)	Speaker, sensor, doorbell, contact, LED 24V	2-core 0.75 sqmm

Conduit Routes - Cheat Sheet

Need	Conduit	Wire	Box depth	MCB
Smart light point	25mm RED	1.5 sqmm	60mm circular	6A
Smart switch (Sonoff)	25mm RED	1.5 sqmm + neutral	65 mm	6A
Dumb light point	25mm RED	1.5 sqmm	50 mm	6A
5A socket	25mm BLUE	2.5 sqmm	50 mm	16A
16A socket	25mm BLUE	2.5 sqmm	50 mm	16A
AC point (20A)	25mm BLUE	4 sqmm	50 mm	20A RCBO
Geyser (20A)	25mm BLUE	2.5 sqmm	50 mm	20A RCBO
Hob 25A	25mm BLUE	4 sqmm	hardwire	25A
Cat6 / network	25mm GREY	Cat6 UTP	50 mm plate	-
Camera (PoE)	25mm GREY	Cat6 UTP	IP67 box	-
Speaker / LV	16mm GREY	2-core 0.75	50 mm	-
LED strip 24V DC	16mm GREY	2-core 0.75	driver only	-

3. Switch-Box Cavity Depth (NEW - critical)

CRITICAL

OLD HABIT: 50 mm GI box at every switch.
NEW REQUIREMENT: 65 mm deep GI box at every SMART-switch location.

If you fit 50 mm boxes at smart-switch locations, the Sonoff / Aqara relay module WILL NOT FIT. The whole installation will need to be redone after plaster - this means breaking the wall, re-chasing, re-plastering, re-painting. DO NOT make this mistake.

The wall plate (front cover) is the same modular size whether the box is 50 mm or 65 mm. Only the WALL HOLE is deeper for the 65 mm version. Buy 65 mm deep GI MS boxes from your hardware shop and physically check the depth before sending to site.

Where 65 mm Deep Boxes Go

Anywhere the switch is marked 'Smart' on the room point schedule. Specifically:

Floor	Location	Gangs	Box size (mm)
GF	Foyer (W wall, near door)	4	230 x 75 x 65
GF	Foyer 2-way at staircase entry	2	130 x 75 x 65
GF	Living Area (W wall)	4	230 x 75 x 65
GF	Living Area 2-way (stair side)	2	130 x 75 x 65
GF	Dining	2	130 x 75 x 65
GF	Master Bedroom (door)	4	230 x 75 x 65
GF	Pooja (outside entry)	2	130 x 75 x 65
GF	Staircase base	1	75 x 75 x 65
FF	FF Living corridor	2	130 x 75 x 65
FF	FF Staircase top (2-way)	1	75 x 75 x 65
FF	Bedroom 1 (inside door)	4	230 x 75 x 65
FF	Bedroom 2 (inside door)	4	230 x 75 x 65
FF	Front balcony (inside)	1	75 x 75 x 65

Where 50 mm STANDARD Boxes Are OK

All sockets (5A, 16A, AC), geyser switches, PIR bathroom switches, kitchen / utility / store room switches (these are dumb - no relay), and Cat6/data wall plates.

4. Universal Smart-Switch Wiring (apply at every smart point)

At every smart-switch location, leave a 300 mm tail of L (red) / N (black) / E (green-yellow) wires inside the box. Cap them with insulated wire connectors. The Sonoff or Aqara module is inserted by the homeowner LATER - you do NOT install the module. You only need to:

1. Use the 65 mm-deep box.
2. Pull L + N + E into the box (3 wires).
3. Pull a switched-live (blue) leg out of the box, up the conduit, to the load.
4. Leave 300 mm tails on all wires.
5. Cap and label.

Schematic

```
                BOX  (75 x 75 x 65 mm GI MS)
               +-------------------------+
FROM CEILING --|  L   N   E              |  <- 3 wires arrive
(power feed)   |   .                     |
               |  +-----------------+    |
               |  |   SONOFF R2     |    |  <- module added LATER
               |  |   Zigbee Relay  |    |     L IN  : Live
               |  |                 |    |     N IN  : Neutral
               |  +--+----------+---+    |     L OUT : to switch
               |     |          |        |     S1/S2 : physical rocker
               |  +--+--+       |        |
               |  |SWITCH|      |        |
               |  |rocker|      |        |
               |  +------+    --+--      |
               |              LOAD  ->   |  <- blue 'switched live' to light
               +-------------------------+
                       |
                  FRONT PLATE
                  (modular cover)
```

5. GF Switch-Board Schedule

Location	Gangs	Box depth	Smart gangs	Notes
Foyer (W wall, near door)	4	65 mm	2 smart + 2 dumb	Lights, screen, 2 spare
Living (W wall)	4	65 mm	3 smart	Main, cove, TV wash, master off
Living 2-way (stair side)	2	65 mm	2 smart	Pairs with W wall
Dining	2	65 mm	1 smart	Pendant + cove
Kitchen (entry)	4	50 mm	0	Lights + exhaust (DUMB)
Utility	1	50 mm	0	Light only (DUMB)
Store room	1	50 mm	0	Single LED (DUMB)
MBR door	4	65 mm	3 smart	Main, reading L+R, master off
GF Bath (outside door)	2	50 mm	PIR auto	PIR module + 20A geyser switch
Staircase base	1	65 mm	Smart 2-way	Pairs with FF landing
Pooja (outside entry)	2	65 mm	2 smart	Ceiling + niche backlight

>> DB panel position

DB recess: 400W x 600H x 100D mm at 1500 mm FFL on West wall, foyer (behind door swing). 48-way Schneider Acti9 IEF48. Pull mains supply from meter, 25 mm conduit, into top of DB.

6. FF Switch-Board Schedule

Location	Gangs	Box depth	Smart gangs	Notes
FF corridor	2	65 mm	2 smart	FF Living lights
FF Staircase top (2-way)	1	65 mm	Smart 2-way	Pairs with GF base
Bedroom 1 (inside door)	4	65 mm	3 smart	Main, reading L+R, master off
BR1 geyser switch (outside T1)	1	50 mm	0	20A DP with neon, 1050 mm
Bedroom 2 (inside door)	4	65 mm	3 smart	Main, reading L+R, master off
BR2 geyser switch (outside T2)	1	50 mm	0	20A DP with neon, 1050 mm
Front balcony (inside)	1	65 mm	Smart	Sunset schedule

NEW Wall Plates on FF (Network)

Location	Type	Height	Notes
FF Living central wall	2x Cat6 keystone wall plate (50 mm	2400 mm	Run R-FF-1 (router/AP)
FF Living beside Cat6	1-gang 5A power socket (50 mm)	2400 mm	Power for AP / PoE injector
BR1 study wall	1x Cat6 keystone (50 mm)	700 mm	Run R-FF-3
BR2 study wall	1x Cat6 keystone (50 mm)	700 mm	Run R-FF-2
FF Front balcony soffit	Cat6 IP67 keystone	soffit	Run R-FF-4 - cap until used

7. Drilling & Cavity Master Reference

Foyer Screen Cavity (most critical - do FIRST)

Cavity W x H x D	540 x 340 x 100 mm
Bottom of cavity (FFL)	1280 mm
Centre of cavity (FFL)	1450 mm (eye level)
Top of cavity (FFL)	1620 mm
Horizontal	Centred on the 1828 mm (6 ft) wall
Conduit entries	2 stubs, bottom-LEFT corner of cavity back wall
Conduit 1 (P-25 RED)	Power - DB to cavity
Conduit 2 (LV-25 GREY)	Cat6 - staircase niche to cavity
LV-16 sleeve	Screen-bezel camera at 1600-1650 mm FFL (top centre)
Inside finish	MATTE BLACK paint inside cavity before stone cladding
VESA backing	12 mm ply, 600 x 400 mm, fixed to cavity back, rawl-plugged

DB Recess

Cavity W x H x D	400 x 600 x 100 mm
Bottom of DB (FFL)	1500 mm
Wall	West wall, foyer, behind door swing
Min clearance from door frame	100 mm

Ceiling Light Holes (use core drill)

Light type	Core hole	Location	Depth
Recessed COB 12W	85 mm core	All bedrooms + living + dining centre	60 mm
Recessed COB 9W	75 mm core	Dining downlights, bath ceiling	60 mm
Recessed GU10 (gimbal)	75 mm core	Foyer spots + bedside reading	65 mm
LED panel 24W kitchen	330 x 330 mm cut	Ceiling centre kitchen	Surface mount
IP65 7W bath downlight	75 mm core	Bathroom shower zone	60 mm
IP65 exhaust fan	165 mm core	Bathroom shower zone corner	thru slab

Niche Cavities (build before plaster/tile)

GF Bath upper niche	600 W x 200 H x 120 D mm (south wall, x=7.25 ft, z=1.4 m)
GF Bath lower niche	400 W x 120 H x 100 D mm (200 mm below upper)
FF T1 niche	600 W x 300 H x 100 D mm (shower south wall)
FF T2 niche	Same spec
Pooja idol niche	TBD - confirm with carpenter; typical 600 W x 900 H x 200 D mm

8. FF Wi-Fi Router / AP Conduits (NEW)

These are NEW conduit runs. They were not in the earlier plan. They go from the staircase niche (GF) up the staircase wall, then branch on FF. They MUST be chased before plastering closes the FF walls.

Run R-FF-1 - Primary FF Router/AP

Conduit	25 mm LV-25 (grey)
Cables	2 x Cat6 UTP + draw wire
Route	Staircase niche -> vertical up staircase W wall -> FF slab level -> branch east along FF Living ceiling/wall -> terminate on FF Living CENTRAL wall, between BR1 and BR2 doors
Termination height	2400 mm FFL (high mount, best signal)
Termination box	50 mm wall plate, 2 x Cat6 keystone
Power	5A socket on D9 circuit, 300 mm to side, same height (2400 mm FFL)

Run R-FF-2 - BR2 study Cat6 drop

Conduit	25 mm LV-25 (grey)
Cables	1 x Cat6 UTP + draw wire
Route	Staircase niche -> vertical -> branch west into BR2 wall -> N wall study area
Termination height	700 mm FFL (study desk level)

Run R-FF-3 - BR1 study Cat6 drop

Conduit	25 mm LV-25 (grey)
Cables	1 x Cat6 UTP + draw wire
Route	Staircase niche -> vertical -> branch east into BR1 wall -> study area
Termination height	700 mm FFL

Run R-FF-4 (optional) - Front Balcony AP future

Conduit	16 mm LV-16 (grey)
Cables	1 x Cat6 outdoor UV + draw wire (cap until needed)
Route	FF Living AP wall plate -> through partition -> front balcony soffit
Termination	Outdoor IP67 keystone box (cap unused)

CRITICAL

STAIRCASE WALL CHASE - cut 150 mm WIDE x 60 mm DEEP chase to fit:
5 x 25 mm power conduits (D circuits + ACs)
3 x 25 mm LV-25 grey (R-FF-1, R-FF-2, R-FF-3)
1 x 16 mm LV-16 grey (existing CAM/sensor runs)
Total: 9 conduits vertical. Cut chase WIDE on first attempt - cutting twice = double labour.

9. Pre-Plaster Sign-Off Checklist

Walk through this checklist with the homeowner BEFORE the mason starts plastering. Once plaster goes on, opening it again is expensive and messy.

[ ] Foyer screen cavity: 540W x 340H x 100D mm at 1280 mm FFL bottom, centred on wall
[ ] DB recess: 400W x 600H x 100D mm at 1500 mm FFL on West wall, foyer
[ ] All smart-switch boxes ARE 65 mm deep (not 50) - physically checked at: Foyer (4-mod 230x75x65), Living main (4-mod), Living 2-way (2-mod), Dining (2-mod), MBR (4-mod), Pooja (2-mod), Staircase base (1-mod), FF Living (2-mod), FF Stair top (1-mod), BR1 (4-mod), BR2 (4-mod), Front balcony (1-mod)
[ ] All dumb switch / socket / geyser switch boxes are 50 mm GI MS
[ ] NEUTRAL wire (black) visible in tail bundle at every smart-switch box
[ ] EARTH wire (green/yellow) visible at every switch and every socket
[ ] All cable tails labelled with masking tape + permanent marker (circuit ID)
[ ] Foyer screen cavity: 2 conduit stubs in bottom-LEFT corner (P-25 power + LV-25 Cat6)
[ ] Foyer screen cavity: LV-16 sleeve to top-centre at 1600-1650 mm FFL (camera)
[ ] Inside foyer screen cavity: matte black paint applied
[ ] Cavity VESA backing: 12 mm ply 600 x 400 mm fixed
[ ] CAM-1 stub (face camera, outside main door, latch side, 1650 mm FFL)
[ ] CAM-2 stub (porch overview, 2400-2700 mm FFL)
[ ] CAM-3 stub (FF front balcony NW corner soffit)
[ ] CAM-4 stub (E-wall exterior at kitchen/utility, 2400-2600 mm FFL)
[ ] CAM-5 stub (terrace staircase parapet, ~2500 mm from terrace floor)
[ ] Doorbell stub (1400-1450 mm FFL, latch side)
[ ] Door contact sensor (top of door frame, concealed)
[ ] Staircase niche: 12-port Cat6 keystone patch panel space at 700 mm FFL (side wall)
[ ] Staircase niche: UPS socket 16A at 300 mm FFL, server sockets 4 x 5A at 400 mm FFL
[ ] RUN R-FF-1: LV-25 stub at FF Living central wall, 2400 mm FFL, 2x Cat6 + draw wire
[ ] RUN R-FF-1 power: 5A socket at FF Living wall, 2400 mm FFL, 300 mm to side of Cat6 plate
[ ] RUN R-FF-2: LV-25 stub at BR2 study wall, 700 mm FFL, 1x Cat6 + draw wire
[ ] RUN R-FF-3: LV-25 stub at BR1 study wall, 700 mm FFL, 1x Cat6 + draw wire
[ ] RUN R-FF-4 (optional): 16mm LV stub at front balcony soffit
[ ] All AC points (E1-E6): 20A socket at 1850 mm FFL on each AC wall
[ ] All geyser conduits: 25 mm BLUE, 2.5 sqmm wire, terminating at geyser height in attic
[ ] Earth bus continuity tested at every metallic body (clamp meter < 1 ohm)
[ ] RCD trip test: simulate 30 mA leakage on each RCBO - must trip in < 30 ms
[ ] PIR switch test (3 bathrooms): walk in, light ON; walk out, wait 10 min, light OFF
[ ] Geyser switch + neon (3 bathrooms): switches ON shows neon, OFF shows no neon
[ ] Photograph EVERY wall before plastering with circuit IDs labelled - keep photos in project folder

10. Common Mistakes To AVOID

>> 50 mm box at smart-switch location

The Sonoff/Aqara module will not fit. Whole installation has to be redone. Always 65 mm at smart-switch locations.

>> Skipping neutral at switch boards

Smart relay needs neutral. Without it, the smart switch is dead the day it's installed. Every switch board gets neutral - even dumb ones, in case we upgrade later.

>> Power and Cat6 in same conduit

Cat6 picks up 230V interference and slows the network. ALWAYS separate conduits.

>> Geyser switch INSIDE the bathroom

Code violation (IS 3043). Always place geyser switch OUTSIDE bathroom door, on the dry side, at 1050 mm FFL.

>> AC socket low on the wall

AC socket at 300 mm puts the cable in view and creates a trip hazard. AC socket is at 1850 mm FFL, near where the indoor unit will mount.

>> Forgetting earth at sockets

Every 3-pin socket gets earth (green/yellow). Test continuity before plastering.

>> Mixing wire colours

Red = LIVE. Black = NEUTRAL. Green/Yellow = EARTH. Blue = SWITCHED LIVE. Never use random colours.

>> Cutting chase narrower than needed

FF staircase chase needs 9 conduits. Cut 150 mm wide first time. Cutting twice = double the mason cost.

>> Foyer screen cavity painted normal colour

Inside cavity must be MATTE BLACK before stone cladding. Otherwise edge shadows show around the screen.

>> Forgetting LV-16 to screen-bezel camera

RPi camera at top centre of foyer screen needs a small CSI ribbon path. Pull 1x LV-16 sleeve from cavity back to top centre at 1600-1650 mm FFL.

11. Sign-Off

By signing below, the electrician confirms that the installation conforms to the layout, cavity depths, conduit colours, wire colours, and standard heights described in this document. Any deviation has been agreed in writing with the homeowner.

Pre-plaster sign-off:

Electrician signature: ______________________ Date: ____________

Mason signature: ______________________ Date: ____________

Homeowner signature: ______________________ Date: ____________

Post-installation sign-off (after fixtures installed):

Electrician signature: ______________________ Date: ____________

Homeowner signature: ______________________ Date: ____________

Document prepared from: electrical/conduits-and-cavities.md, ground-floor-electrical.md, first-floor-electrical.md, db-layout.md, materials-checklist.md (project repo). For digital reference, see the markdown source files. v1.0 - 2026-05-06.